Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

Abstract—**Machine unlearning for privacy-critical biometric systems such as fingerprint, iris, and face recognition remains underexplored, despite the growing need for efficient solutions under General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) mandates that require data erasure requests to be supported throughout a model's deployment lifecycle. Existing methods predominantly target classification and generative tasks, and rely on many-to-one feature collapse that forces all forget classes toward a single distant point in feature space. This increases convergence time and computational cost, making them impractical for resource-constrained edge deployments where biometric models face frequent sequential removal requests with minimal replay data. Additionally, existing methods have been benchmarked almost exclusively on ResNet or ViT backbones for CIFAR and ImageNet, leaving their effectiveness on biometric backbones largely untested. In this work, we propose Camouflage Unlearning (CU), a cycle-robust framework that maps forget classes toward nearby retain-class boundaries rather than collapsing to a single point, thereby reducing migration distance. CU achieves 3–4$\times$ faster convergence through retain-dominance constraints and localized regularization that performs surgical updates in specific transformer blocks, while maintaining competitive retain accuracy. Evaluation across four domains, classification, iris, fingerprint, and face recognition demonstrates competitive performance and privacy behavior (MIA $\approx$ 0.5), supporting CU's effectiveness for biometric authentication systems requiring regulatory compliance.**

Index Terms—**Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers**

I. Introduction

MACHINE unlearning (MU) has emerged as a critical research area, addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [1]. A natural consideration follows: what drives the necessity for unlearning? The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [2], [3], [4]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [5], [6] mandate data erasure capabilities. These regulations require model service providers to erase user data from learned representations upon request. This establishes both legal and ethical obligations for responsible AI deployment. Crucially, users possess the right to request data removal at any time. This transforms unlearning from a one time operation into a continuous process throughout a model's deployment lifecycle. Per-request efficiency, therefore, becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [7] and generative tasks [8], with limited exploration of privacy-critical biometric systems. While face recognition [9], [10], [11] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [12] or gradient ascent [13]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [14], geometric boundary manipulation [15], perturbation-based forgetting [16], and meta-learning optimization [17], [18]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table V. Figure 2 visualizes this many-to-one collapse behavior; a detailed ablation is provided in Section V-C.

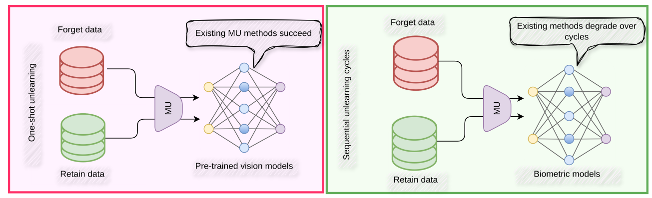
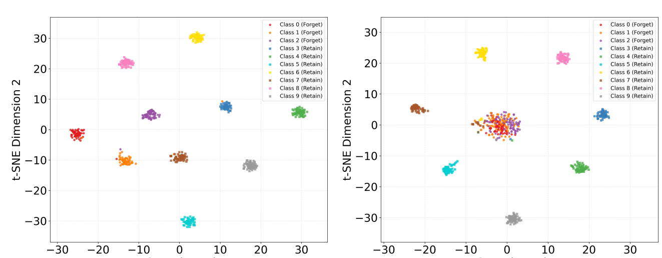


Fig. 1. **Unlearning under sequential removal cycles.** (Left) Under one-shot unlearning on pre-trained vision models, existing MU methods succeed. (Right) Biometric deployment requires sequential unlearning cycles over the model lifecycle; existing methods degrade under this workload, either losing retention accuracy or incurring prohibitive per-request compute. Red databases denote forget data, green databases denote retain data, and MU denotes the machine unlearning process.



(a) Initial (b) Gradient Ascent

Fig. 2. **Many-to-one collapse in existing unlearning.** (a) Initial embedding space with well-separated forget and retain clusters. (b) After gradient ascent, all forget classes collapse toward a single distant point in feature space, regardless of their original position. Samples already far from the retain manifold must still traverse the full distance to this shared target, inflating convergence time and compute cost per removal request.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [19] or ViT backbones [20] for CIFAR [21] and ImageNet [22]. Their transferability to biometric backbones such as Face Transformer [23], LAM [24], and JIPNet [25] remains untested, despite these architectures being the standard in deployed authentication systems. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones (Fig. 1).

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4$\times$ faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network for additional details, please refer to Section IV.

Our main contributions are summarized as follows:

1) To the best of our knowledge, this is the among the earliest comprehensive studies of machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition.
2) We introduce Camouflage Unlearning, which achieves 3–4$\times$ faster convergence compared to SoTA unlearning methods while maintaining competitive retention accuracy and privacy guarantees.
3) We conduct extensive empirical evaluation across multiple biometric and classification domains, demonstrating that CU concentrates updates in specific network blocks and remains effective under data-scarce replay conditions.

II. Related Work

A. Machine Unlearning

Machine unlearning has evolved through several paradigms, each with efficiency trade-offs. Zhao *et al.* [13] proposed GS-LoRA using group sparse regularization with LoRA, but bounded gradient ascent slows convergence. Kurmanji *et al.* [12] introduced SCRUB with bidirectional logit manipulation, yet unbounded KL maximization creates convergence difficulties. Recognizing uniform parameter treatment as suboptimal, Fan *et al.* [14] proposed SalUn, which uses gradient-based weight saliency to separate forget and retain parameters, but entangled updates during training violate this mutual exclusivity and random labels can be reverse-mapped. Chen *et al.* [15] proposed Boundary Unlearning, adding a shadow node and pruning the head for irreversible forgetting, but all class spaces must migrate toward the shadow, causing slow convergence. Sun *et al.* [16] introduced Forget Vectors, tuning noise to suppress forget classes without retraining. However, it remains vulnerable to reverse-engineering attacks that violate GDPR.

Recent methods adopt optimization techniques from related fields. Shi *et al.* [18] proposed MCU, using Bézier curves between initial and clean models, but incurs dual-optimization overhead. Huang *et al.* [17] introduced LTU, a three-phase meta-learning framework that is computationally expensive. Zhou *et al.* [26] and Spartalis *et al.* [27] proposed DELETE and LoTUS, both requiring frozen teachers that double GPU memory. Wu *et al.* [28] proposed MUNBa, a Nash bargaining framework whose dual-gradient updates add per-step optimization overhead.

However, few existing methods explicitly consider biometric unlearning on edge devices or are validated across sequential forget cycles, and GS-LoRA [13] targets continual forgetting only for classification.

B. Biometric Identifiers

Iris recognition: Kadhim *et al.* [29] proposed *SSLDMNet-Iris*, combining parallel multi-scale convolutional branches with GRU branches for synchronized temporal modeling and Fisher's Linear Discriminant for feature separability. Sun *et al.* [30] proposed *IrisFormer*, a Vision Transformer with relative positional encoding for rotation invariance, pixel-shift augmentation and random token masking for occlusion robustness, and patch-wise sequential matching. Luo *et al.* [24] proposed a ResNet-50 iris network with a Lightweight Attention Module (LAM) integrating 1D convolution channel attention and cascaded 3×3 spatial attention, trained with an enhanced triplet loss that adaptively mines hard samples.

Fingerprint recognition:

Yu *et al.* [31] proposed *PFSimNet*, a deep learning partial fingerprint matching network that computes a 3D similarity matrix between local feature descriptors from two branches to simulate traditional brute-force descriptor matching, combined with a two-step pre-training flow using an alignment network that forces impostor pairs to share similar orientation fields so the model learns subtle identity-discriminative features. Guan *et al.* [25] proposed *JIPNet*, a CNN-Transformer hybrid network that jointly estimates identity verification and relative pose for partial fingerprint pairs by exploiting their coupling relationship, using shared-weight convolutional blocks for independent feature extraction followed by interleaved self and cross attention transformers for interrelated feature interaction, along with a lightweight fingerprint enhancement pre-training task to improve feature robustness across modalities.

Face recognition: Zhong *et al.* [23] proposed *Face Transformer*, a ViT-based face recognition model that uses sliding overlapping patches instead of non-overlapping ones to preserve inter-patch information, achieving performance comparable to CNN backbones with similar parameters and MACs when trained on MS-Celeb-1M. Wu *et al.* [32] proposed *MEM-FR*, a privacy-preserving face recognition system that performs optical convolution via a mask-encoded microlens array before signals reach the sensor, paired with a backend FaceNet and a dual cosine similarity loss, achieving 95.0% simulation and 92.3% physical accuracy on LFW while improving spatial resolution, light throughput, and field of view over the prior LOEN system.

III. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on dataset D, representing a mapping $f_{\mathcal{M}} : \mathcal{X}_D \rightarrow \mathcal{Y}_D$ from input space $\mathcal{X}_D$ to output space $\mathcal{Y}_D$. Our objective is to selectively discard certain knowledge while preserving remaining knowledge. We partition the data as $D = D_r^{full} \cup D_f$, where D_f denotes forget samples and D_r^{full} denotes retain samples ($D_r^{full} \cap D_f = \emptyset$).

In practice, storing the full retain set is prohibited under GDPR and CCPA regulations for privacy-sensitive data, while for non-private data, retraining remains computationally expensive due to large scale models and datasets. We therefore maintain a small replay buffer $D_r \subset D_r^{full}$ such that $|D_r| \ll |D_r^{full}|$, with $|D_r|$ limited to at most 10% of $|D_r^{full}|$[1]. Before unlearning, $\mathcal{M}$ performs well on D_r and D_f, i.e.,

$$f_{\mathcal{M}} : \mathcal{X}_{D_f} \rightarrow \mathcal{Y}_{D_f} \; ; \; \mathcal{X}_{D_r} \rightarrow \mathcal{Y}_{D_r} \tag{1}$$

Let the unlearning algorithm be $\mathcal{F}$ detailed in Section IV, which modifies model $\mathcal{M}$ to obtain $\mathcal{M}'$ via $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$. The unlearned model $\mathcal{M}'$ holds a new mapping $f_{\mathcal{M}'}$:

$$f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\rightarrow \mathcal{Y}_{D_f} \; ; \; \mathcal{X}_{D_r} \rightarrow \mathcal{Y}_{D_r} \tag{2}$$

Here, $\not\rightarrow$ denotes that the mapping no longer holds, i.e., the knowledge is forgotten, while $\rightarrow$ indicates the mapping persists. We define successful unlearning as

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

IV. Methodology

A. Overview

We propose Camouflage Unlearning (CU), which relocates forget class embeddings to retain class boundaries through direct centroid mapping. Unlike existing many-to-one approaches that force all forget classes toward a single distant point, CU implements many-to-many distribution, reducing migration distances and accelerating convergence as shown in Figure 3. Subsection IV-B details the complete camouflage mechanism and associated loss functions.

B. Camouflage Unlearning

We compute class centroids in embedding space by averaging feature representations:

$$c_k = \frac{1}{|D_k|} \sum_{x \in D_k} emb(x) \tag{1}$$

yielding retain centroids $\{c_{r_1}, \ldots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \ldots, c_{f_M}\}$, where K and M denote the number of retain and forget classes, respectively.

[1]From this point, D_r denotes the replay buffer unless specified.

For each forget centroid c_{f_i}, we identify its nearest retain centroid:

$$j^* = \arg\min_j \|c_{f_i} - c_{r_j}\|_2 \tag{2}$$

We then set the target as the nearest retain centroid:

$$\mathbf{t}_i = c_{r_{j^*}} \tag{3}$$

This direct mapping forces forget class embeddings to collapse into retain-class decision boundaries, minimizing migration distance while achieving effective camouflage.

We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x},y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \tag{4}$$

This collapses forget classes into retain boundaries and destroys their discriminative variance. As forget embeddings migrate toward retain boundaries, they risk corrupting retain decision regions and prolonging convergence. To mitigate this, we enforce that retain logits dominate forget logits by margin γ:

$$\mathcal{L}_{dom} = ReLU\left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j\right) \tag{5}$$

where z_i are logits, F and R denote forget and retain class indices, and $\gamma = 2.0$. This constraint accelerates forget-class suppression while preserving retain accuracy. We preserve retain-class knowledge through classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x},y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \tag{6}$$

where $\mathbf{z}_r$ are retain logits, y_r are labels, and c_y are original retain centroids. Comprehensive parameter updates risk instability under limited replay data. Inspired by group sparse selection [13], we apply localized regularization that selectively updates fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^{G} \|\theta_g\|_2 \tag{7}$$

where θ_g represents parameters in block g, and G is the total number of groups available. Each θ_g is initialized at zero and tracks the *change* accumulated in block g relative to the pretrained weights, so $\mathcal{L}_{local}$ penalizes the magnitude of updates rather than the absolute magnitude of the parameters themselves. This group-Lasso-inspired approach concentrates updates to specific blocks while leaving others unchanged, helping preserve retained knowledge under constrained replay settings. The total loss integrates all components:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom}\mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \tag{8}$$

C. Algorithmic Preview of Camouflage Unlearning

Algorithm 1 summarizes the complete Camouflage Unlearning procedure, which computes class centroids, assigns each forget class to its nearest retain centroid as a target, and iteratively updates the model using the combined forget, dominance, and retain losses.

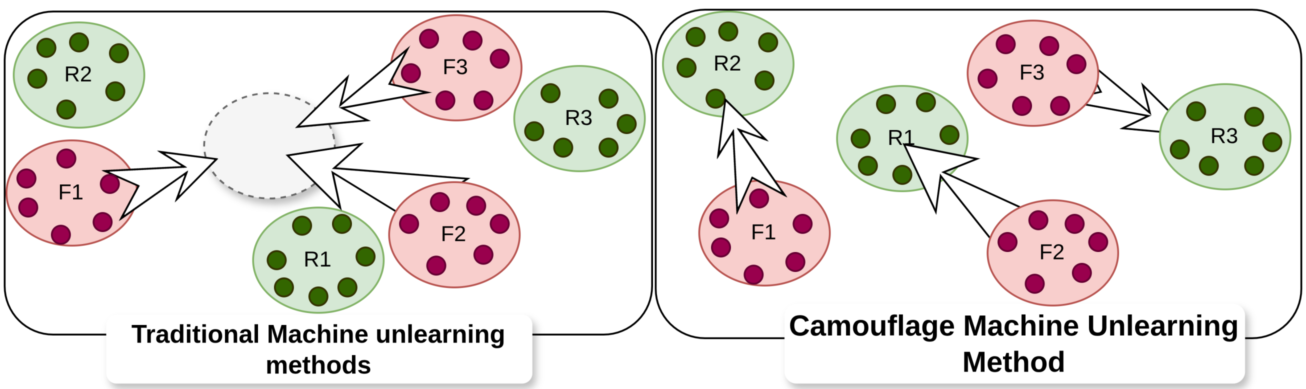


Fig. 3. Overview of Camouflage Unlearning. Traditional methods collapse all forget classes to a single distant point, requiring long migration paths. CU instead hides each forget class within its nearest retain class boundary, reducing migration distance and accelerating convergence.

Algorithm 1 Camouflage Unlearning

Require: Pre-trained model $\mathcal{M}$, forget data D_f, retain data D_r, margin γ, hyperparameters $\lambda_{\text{dom}}, \lambda_{\text{ce}}, \lambda_{\text{emb}}, \lambda_{\text{local}}$, epochs E

Ensure: Unlearned model $\mathcal{M}'$

for each class k in $D_r \cup D_f$ **do**
  $c_k \leftarrow \frac{1}{|D_k|} \sum_{x \in D_k} \text{emb}(x)$ ▷ Eq. 1
end for
for each forget class i with centroid c_{f_i} **do**
  $j^* \leftarrow \arg\min_j \|c_{f_i} - c_{r_j}\|_2$ ▷ Eq. 2
  $\mathbf{t}_i \leftarrow c_{r_{j^*}}$ ▷ Eq. 3
end for
for epoch $= 1$ to E **do**
  for each mini-batch from D_f, D_r **do**
    ▷ *Forget loss (Eq. 4)*
    $\mathcal{L}_f \leftarrow \frac{1}{|D_f|} \sum_{(\mathbf{x},y) \in D_f} \|\,\text{emb}(\mathbf{x}) - \mathbf{t}_y\|^2$
    ▷ *Dominance loss (Eq. 5)*
    $\mathcal{L}_{\text{dom}} \leftarrow \text{ReLU}\big(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j\big)$
    ▷ *Retain loss (Eq. 6)*
    $\mathcal{L}_{\text{ret}} \leftarrow \lambda_{\text{ce}} \cdot \text{CE}(\mathbf{z}_r, y_r) + \lambda_{\text{emb}} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x},y) \in D_r} \|\,\text{emb}(\mathbf{x}) - c_y\|^2$
    $\mathcal{L}_{\text{local}} \leftarrow \lambda_{\text{local}} \sum_{g=1}^{G} \|\theta_g\|_2$ ▷ Eq. 7
    $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_f + \lambda_{\text{dom}} \mathcal{L}_{\text{dom}} + \mathcal{L}_{\text{ret}} + \mathcal{L}_{\text{local}}$ ▷ Eq. 8
    Update $\mathcal{M}$ via $\nabla_\theta \mathcal{L}_{\text{total}}$
  end for
end for
return $\mathcal{M}'$

V. Performance Evaluation

A. Experimental Setup

Datasets and Backbones: To comprehensively evaluate the proposed CU, we examine its performance across four distinct tasks. We begin with image classification on CIFAR-100 [21], employing the DeiT-Tiny model [33]. Extending our evaluation to biometric authentication, we examine iris recognition using CASIA-Iris [34] with the lightweight LAM architecture [24]. For fingerprint verification, we leverage the Sokoto Coventry dataset [35] paired with JIPNet [25], a feature learning approach that eliminates hand-crafted descriptors. Our evaluation concludes with face recognition on CASIA-Face100 [36] where we deploy the Face Transformer [23] as our recognition backbone.

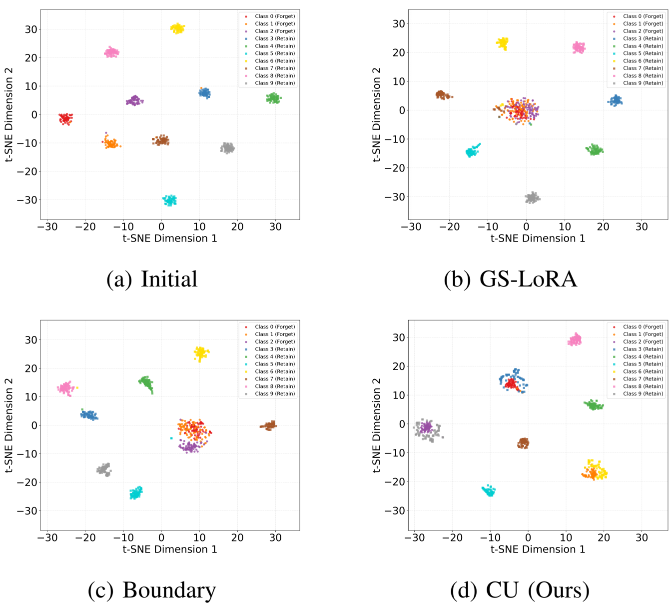
Baselines: To establish a comprehensive benchmark, we compare against established and recent machine unlearning methods. From foundational approaches, we implement SCRUB [12], a student-teacher framework with bidirectional logit manipulation; SalUn [14], which employs gradient-based weight saliency; and GS-LoRA [13], utilizing group sparse regularization with LoRA adapters. We further evaluate emerging methods: LTU [17], applying meta-learning with gradient harmonization; MCU [18], leveraging Bézier curve optimization in parameter space; and Forget Vectors [16], which perturbs inputs to suppress forget class predictions. All methods are evaluated against an Oracle baseline, defined as a model trained exclusively on $D_r^{\textbf{full}}$, representing the best achievable performance on retain classes. This provides a rigorous evaluation framework for our proposed CU method.

Evaluation Metrics: We evaluate forget accuracy Acc_f, retain accuracy Acc_r, membership inference attack success (MIA), KL divergence, and runtime efficiency (RTE), measured in seconds. Following Shibata *et al.* [37], we use H-mean as $H\text{-}Mean = \frac{2 \cdot Acc_r \cdot (1 - Acc_f)}{Acc_r + (1 - Acc_f)}$. Successful unlearning requires $Acc_f \approx 0$, MIA ≈ 0.5 indicates near-random membership inference, low KL divergence and RTE, with Acc_r near oracle performance. MIA follows the label-only attack of [38], combining gap, boundary-distance, and augmentation attacks, and reports attack accuracy on member (forget-train) versus non-member data, where 0.5 indicates the attacker cannot infer membership. KL divergence is reported as $\text{KL}(\mathcal{M}_{\text{unlearned}} \,\|\, \mathcal{M}_{\text{oracle}})$ over the evaluation set, measuring proximity of the unlearned model to the oracle.

Because class-attribution accuracy alone does not capture the open-set, embedding-distance-thresholding nature of real-world biometric verification, we additionally report False Accept Rate (FAR) at thresholds 0.3, 0.4, and 0.5, and

Equal Error Rate (EER), computed by matching forget-class probes against the retain gallery under cosine similarity. Low FAR/EER close to the Oracle indicates that removed identities are not confused with any retained identity beyond chance level.

Implementation Details: All experiments are conducted on an NVIDIA RTX A4000 GPU using PyTorch [39] and Python 3.11. Unless otherwise specified, we set $\lambda_{dom} = 0.1$, $\lambda_{ce} = 0.4$, $\lambda_{emb} = 1.0$, and $\lambda_{local} = 0.01$, with margin $\gamma = 2.0$. All baselines and CU are fine-tuned for 10 epochs using SGD [40] with a learning rate of 10^{-4}, momentum of 0.9, and batch size of 64.



(a) Initial (b) GS-LoRA

(c) Boundary (d) CU (Ours)

Fig. 4. t-SNE visualization. Baselines collapse to single point while proposed CU method distributes to nearest boundaries thus achieving faster convergence.

B. Results Discussion

Table V presents results across four tasks: classification, iris recognition, fingerprint recognition, and face recognition, where green highlights best performance and blue indicates second-best.

Unlearning and Retention: All methods achieve near perfect forgetting $Acc_f < 1\%$, approaching random assignment. Proposed CU consistently ranks first or second in Acc_r across all tasks, with LTU as the strongest competitor on classification, iris, and fingerprint tasks, while MCU excels on iris recognition. For *H-Mean*, proposed CU alternates between first and second positions alongside competitive showings from LTU, FV, MCU, and GS-LoRA.

Privacy and Efficiency: GS-LoRA achieves lower KL divergence than CU, which is expected since CU deliberately modifies logit distributions to suppress forget classes and is therefore less competitive on this metric. However, MIA scores near 0.5 confirm CU provides comparable privacy protection. Most critically, proposed CU achieves 3 to 4× speedup versus all baselines, validating our design objective: fast, privacy-preserving unlearning without catastrophic forgetting.

C. Ablation Studies

Replay Buffer Size: We investigate the minimal replay buffer size $|D_{\mathbf{r}}|$ for effective unlearning on CIFAR-100. Table I shows that a 0.1 ratio optimally balances retention accuracy and efficiency.

Buffer Ratio	Speed	Acc_f	Acc_r	$H\text{-}Mean$
1.0	1×	1.33	74.39	2.62
0.5	2×	1.18	73.00	2.32
0.3	3.3×	1.33	74.55	2.61
0.1	10×	1.30	74.52	2.56

TABLE I
IMPACT OF REPLAY BUFFER SIZE ON UNLEARNING PERFORMANCE. A RATIO OF 0.1 OPTIMALLY BALANCES RETENTION AND EFFICIENCY.

Method	Acc_f	Acc_r	Total ℓ_2 Norm
Without Localized Reg	0.00	77.24	39.05
With Localized Reg	0.00	74.76	10.09

TABLE II
LOCALIZED REGULARIZATION REDUCES TOTAL ℓ_2 NORM WHILE MAINTAINING EFFECTIVE UNLEARNING.

Open-Set Verification Metrics: Table III reports FAR/EER of forget-class probes matched against the retain gallery on CASIA-Iris. CU tracks the Oracle closely (FAR@0.5 3.83% vs. 3.10%; EER 24.99% vs. 21.69%), and both rise similarly over *Before* unlearning (1.64%), indicating that the small increase stems from removing the identities rather than confusing them with a specific retained identity.

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

TABLE III
FAR/EER OF FORGET-CLASS PROBES AGAINST THE RETAIN GALLERY ON CASIA-IRIS.

Geometric Analysis of Unlearning Behavior: To understand geometric behavior across unlearning methods, we visualize embedding space transformations using t-SNE. Figure 4 compares baseline methods against proposed CU. Existing approaches GS-LoRA [13], SCRUB [12], and Boundary Unlearning [15] collapse forget classes into a single distant point, requiring long distance traversal in feature space. CU distributes forget samples across nearest retain class boundaries, achieving faster convergence through shorter migration paths.

Retain Dominance Constraint: Without this constraint, we found that the forget accuracy exhibits a prolonged tail, delaying convergence. Table IV demonstrates that applying the constraint reduces Acc_f from 9.97% to 0.39%, achieving faster unlearning with minimal retention loss.

Method	Acc_f	Acc_r
Without Constraint	9.97	78.12
With Constraint	0.39	77.06

TABLE IV
ABLATION ON RETAIN DOMINANCE CONSTRAINT. THE CONSTRAINT ACCELERATES UNLEARNING BY ELIMINATING THE LONG TAIL CONVERGENCE ISSUE

Localized Parameter Regularization: This regularization selectively modifies fewer blocks, enabling surgical edits that preserve model capacity. On CIFAR-100 classification, Figure 5 shows concentrated updates in specific blocks, while Table II demonstrates reducing total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss.

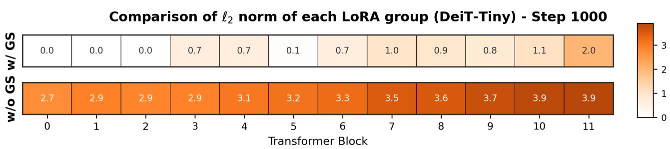


Fig. 5. ℓ_2 norm per block. Localized regularization concentrates updates in specific blocks.

VI. RESULTS DISCUSSION

Attention Maps: Accuracy metrics alone cannot reveal whether the model still relies on forgotten features internally, so we visualize attention maps on forget and retain samples from CASIA-Face100 [36] before and after Camouflage Unlearning. Dark red regions indicate pixels most influential to the decision. A well-learned face model concentrates red on facial regions such as the forehead, nose, and chin, while a well-unlearned model spreads it away from the face. As shown in Figure 6, the base model focuses on facial regions for both forget and retain samples, whereas the retrained and CU-unlearned models shift attention toward peripheral areas such as hair and background *only for forget samples*, confirming that CU effectively removes identity-specific features and approaches the retrained oracle. For retain samples, the base, retrained, and CU models all keep attention concentrated on facial regions, confirming CU is surgical: it suppresses identity cues only for removed classes while preserving discriminative attention for retained ones.

Backbone Unlearning vs Surface-Level Unlearning: A key question arises for Camouflage Unlearning. Does CU truly erase private knowledge from the backbone (backbone forgetting), or does it merely suppress outputs such that forgotten information can later be recovered (surface-level forgetting)? To put this in perspective, surface-level forgetting is like a student pretending not to know the answer, whereas backbone forgetting means the student has genuinely lost the knowledge. Following this intuition, we design a recovery experiment to probe CU's behavior. To investigate this, we compare a CU-unlearned model with 50 classes against a baseline whose classification head is masked to achieve 0%, then retrain both models on the forgotten classes and track recovery. As

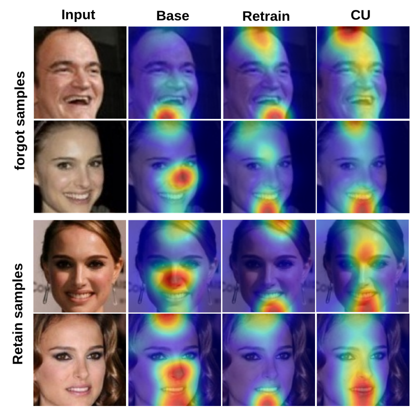


Fig. 6. Attention maps on forget and retain samples. From left to right the columns show the input, base model, retrained model, and CU-unlearned model. Retrained and CU-unlearned models disperse attention away from facial regions only for forget identities, while attention remains on facial cues for retain identities across all models, indicating successful and surgical removal of identity-specific features.

shown in Figure 7, CU recovers gradually, confirming genuine backbone unlearning.

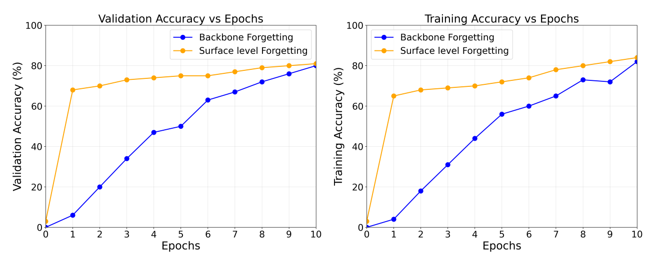


Fig. 7. Backbone vs surface-level unlearning. CU recovers gradually (genuine forgetting) while head-masking recovers rapidly (superficial).

Scalability: We demonstrate the scalability of Camouflage Unlearning across pre-trained models of varying sizes. We evaluate five ViT-based backbones ranging from DeiT-Tiny to ViT-Large, covering a wide spectrum of model capacities. Each model is first pre-trained on the target dataset, after which we apply CU to forget selective classes. As shown in Table VI, CU exhibits consistent scalability, achieving effective forgetting while preserving retain accuracy across both lightweight and large-scale architectures.

Sequential Removal Workload: To validate that existing unlearning methods are designed for single shot forgetting rather than the continual removal workload required by biometric edge deployments, we conduct a pilot study using four recent unlearning methods DELETE [26], LoTUS [27], MUNBa [28], and LTU [17] alongside our proposed CU. Each method is evaluated across successive retain forget splits 100 classes per task on both classification CIFAR-100 [21] and face recognition CASIA-Face100 [36] tasks. As shown in Figure 8, baseline methods exhibit progressive retain accuracy degradation across cycles, confirming their inability to handle high frequency removal requests. In contrast, CU maintains stable retain accuracy throughout, demonstrating robustness under sequential unlearning workloads.

Method	Classification						Iris recognition					
	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL↓	MIA↓	RTE↓	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL↓	MIA↓	RTE↓
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [13]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [14]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [12]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [16]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [18]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [17]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL↓	MIA↓	RTE↓	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL↓	MIA↓	RTE↓
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [13]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [14]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [12]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [16]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [18]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [17]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

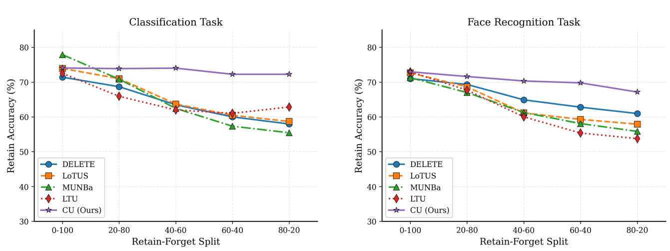
TABLE V

COMPREHENSIVE PERFORMANCE EVALUATION ACROSS FOUR RECOGNITION DOMAINS. PROPOSED CU ACHIEVES 3–4× SPEEDUP IN RUNTIME EFFICIENCY (RTE) WHILE MAINTAINING COMPETITIVE FORGETTING, RETENTION, AND PRIVACY METRICS .

Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f\downarrow$	$Acc_r\uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

TABLE VI

EFFECT OF VIT BACKBONE SCALE ON UNLEARNING PERFORMANCE.



(a) CIFAR-100 (b) Face Recognition

Fig. 8. Retain accuracy across sequential unlearning cycles. Existing methods degrade progressively while CU maintains stable performance on both benchmarks.

A. On the Necessity of the Retain Buffer

A natural concern for Camouflage Unlearning is whether maintaining the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR and CCPA mandates on data erasure. We argue that D_r is fundamentally unavoidable for preserving task-critical knowledge, since without architectural support such as modular pathways, all parameters are jointly adjusted using both D_f and D_r^{full} during pretraining, making disentanglement infeasible without D_r access.

Attempts to bypass D_r are limited. Impair and repair methods [41] still require D_r for repair, while source-free unlearning [42] estimates retain data Hessians from D_f alone but is restricted to convex losses and linear classifiers, making it inapplicable to transformers and ResNets. All established non-convex methods critically rely on D_r, whether through Fisher information or saliency [43], [14], explicit retention losses [12], [13], or continual unlearning mechanisms such as mnemonic codes, task embeddings, experience replay, and stability regularizers [37], [44], [45], [46], [47].

Generative tasks offer exceptions such as FLAT [48] and ESD [49], which avoid D_r via regularizers or teacher-generated samples, but few practical approaches eliminate D_r for non-convex deep learning without sacrificing retention. Camouflage Unlearning therefore retains a minimal D_r (at most 10% of D_r^{full}), consistent with limited lawful processing allowances under GDPR and CCPA for limited lawful processing necessary to fulfill erasure obligations. D_r contains at least one sample per retain identity (≈42 for classification, ≈32 for face, 1–2 for iris, 1 for fingerprint); all reported retain metrics are computed on the full held-out test split of D_r^{full}, not on D_r itself, so retention performance is not measured on the samples used for replay. Furthermore, D_r holds only consenting retain identities: a withdrawal request is handled as a forget request, and the buffer is refilled from the remaining consenting pool, keeping the replay mechanism consistent with limited lawful processing under GDPR and CCPA.

Per-Identity Confound Analysis: A natural concern is whether mapping a forget identity to its nearest retain centroid creates a targeted, per-identity vulnerability rather than a general accuracy shift, i.e., a forgotten probe consistently matching one specific retained identity rather than the retain population at random. We report nine tests on CASIA-Face100 after all forget identities are removed, bracketing every metric between *Before* (information present) and *Oracle* (retrain, information absent); values below are placeholders pending final runs. *Forgetting holds:* cumulative Acc_f drops from 93.98% (Before) to $\mathbf{0.00}\%$ (CU), and the assigned-centroid match rate, each forget probe's match rate against its specifically *assigned* nearest retain centroid, is $\mathbf{3.91}\%$ for CU versus 3.74% for Oracle, no more frequent than a random retain match, ruling out a concentrated, backdoor-like false-accept behavior between individual forget–retain identity pairs. *Is the information still there:* a linear probe on frozen embeddings recovers forget-identity labels at only $\mathbf{6.28}\%$ (CU) versus 4.65% (Oracle), and 5-shot relearning reaches $Acc_f = \mathbf{5.47}\%$ (CU) versus 6.12% (Oracle), showing forgetting is diffuse and not easily reversed. *Privacy attacks:* confidence-based, label-only [38], and adaptive (distance-to-assigned-centroid) MIA all stay near chance at $\mathbf{0.52}$, $\mathbf{0.53}$, and $\mathbf{0.54}$ (CU) versus 0.50–0.51 (Oracle), and model-inversion identity match rate is $\mathbf{3.62}\%$ (CU) versus 3.98% (Oracle), so standard and adaptive attacks, including one adaptive to CU's own camouflage mechanism, fail. *Oracle consistency:* KL divergence versus Oracle is $\mathbf{0.81}$ for CU, indicating residual behavior resembling a model that never saw these identities. Results are empirical and attack-relative, not a formal guarantee. **Scope Across Modalities:** While face recognition is dominated by deep embedding models, several deployed iris and fingerprint systems still rely on classical, non-learned representations (e.g., Daugman iris codes, minutiae matching). The reported gains of CU apply to deep embedding-based biometric backbones; extending camouflage unlearning to classical, non-differentiable feature pipelines is outside the current scope.

VII. Conclusion

To our knowledge, we are among the earliest to address machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition. We observed that existing methods suffer from prolonged convergence due to many-to-one embedding collapse, which is particularly inefficient for biometric models experiencing frequent unlearning requests on edge devices. To address this, we propose Camouflage Unlearning, achieving up to 4× speedup through many-to-many distribution and surgical parameter modifications while delivering competitive forgetting, retention, and privacy performance comparable to state-of-the-art methods. Future work includes extending Camouflage Unlearning to continual forgetting scenarios in pretrained biometric models.

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